

# JavaScript Random

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## Math.random()

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`Math.random()` returns a random number between 0 (inclusive) and 1 (exclusive):

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Random</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Math.random()</h4>
<p id="demo"></p>

<script>
let text = "Random numbers between 0 and 1:<br><br>";

for (let i = 0; i < 5; i++) {
  text += Math.random() + "<br>";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

## Example 1

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Result [View Example](#)

## JavaScript Random Integers

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`Math.random()` used with `Math.floor()` generates random integers:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Random</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Random Integers</h4>
<p id="demo"></p>

<script>
let text = "Random integers:<br><br>";

// 0 to 9
text += "Math.floor(Math.random() * 10): ";
for (let i = 0; i < 3; i++) {
    text += Math.floor(Math.random() * 10) + " ";
}
text += "<br>";

// 1 to 10
text += "Math.floor(Math.random() * 10) + 1: ";
for (let i = 0; i < 3; i++) {
    text += Math.floor(Math.random() * 10) + 1 + " ";
}
text += "<br>";

// 0 to 100
text += "Math.floor(Math.random() * 100): ";
for (let i = 0; i < 3; i++) {
    text += Math.floor(Math.random() * 100) + " ";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

## Example 2

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Result [View Example](#)

## Random Integers in a Range

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You can generate random integers in a specific range (min to max):

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Random</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Random Integers in a Range</h4>
<p id="demo"></p>

<script>
// Random integer between min (inclusive) and max (inclusive)
function getRndInteger(min, max) {
  return Math.floor(Math.random() * (max - min + 1)) + min;
}

let text = "Random integers between 1 and 6 (dice):<br>";
for (let i = 0; i < 6; i++) {
  text += getRndInteger(1, 6) + " ";
}
text += "<br><br>";

text += "Random integers between 10 and 20:<br>";
for (let i = 0; i < 6; i++) {
  text += getRndInteger(10, 20) + " ";
}

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

## Example 3

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Result [View Example](#)

## A Proper Random Function

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A reusable function for generating random integers between any two values:

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Random</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>A Proper Random Function</h4>
<p id="demo"></p>

<button onclick="rollDice()">Roll Dice</button>
<p id="diceResult"></p>

<script>
// Proper random function: min (inclusive) to max (inclusive)
function getRndInteger(min, max) {
    return Math.floor(Math.random() * (max - min + 1)) + min;
}

function rollDice() {
    const dice1 = getRndInteger(1, 6);
    const dice2 = getRndInteger(1, 6);
    document.getElementById("diceResult").innerHTML =
        "Dice 1: " + dice1 + " | Dice 2: " + dice2 + " | Total: " + (dice1 + dice2);
}

document.getElementById("demo").innerHTML =
    "This function returns a random integer:<br>" +
    "from min (inclusive) to max (inclusive):<br><br>" +
    "function getRndInteger(min, max) {<br>" +
    "    return Math.floor(Math.random() * (max - min + 1)) + min;<br>" +
    "}";
</script>

</body>
</html>

```

## Example 4

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Result [View Example](#)

## Document

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Document in project

You can [Download PDF](#) file.

## Reference

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- [W3Schools JavaScript Random](#)